

Intersection of a Line and a Plane in 3 space. ⁽¹⁾

CASE 1: The line is parallel and distinct from the plane. (no intersection
no solution)

CASE 2: The line intersects the plane in a single point. (1 solution)

CASE 3: The line lies entirely in the plane. (intersection consists of all points on the line. An infinite # of solutions.)

Rule: To find the intersection of a line with a plane in 3-space, substitute the parametric equations of the line into the scalar equation of the plane and solve for the parameter. The co-ordinates of the point of intersection can then be calculated using the value of the parameter.

CASE 1: $l: x = 1 + 3t, y = 2 - 2t, z = 4t$

$\pi: 2x + 5y + z - 4 = 0$

Solution: note: use the dot product to check that they are parallel.

ie: $[3, -2, 4] \cdot [2, 5, 1] = 0$

$\Rightarrow$ parallel.

Point $(1 + 3t, 2 - 2t, 4t)$ lies on the plane if it satisfies the equation of the plane.

$$2(1 + 3t) + 5(2 - 2t) + 4t - 4 = 0$$

$$2 + 6t + 10 - 10t + 4t - 4 = 0$$

$$0t = -8$$

$$0 = -8$$

$\Rightarrow$ no solution

$\therefore$ line and plane are parallel and distinct.

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CASE 2: $l: \vec{r} = [2, -5, 6] + t[1, 3, 5]$

$\pi: 3x - 4y + 2z - 3 = 0$

Solution

note: $[1, 3, 5] \cdot [3, -4, 2]$

$= 3 - 12 + 10 = 1 \neq 0$

$\Rightarrow$ not parallel.

Point $(2+t, -5+3t, 6+5t)$ must lie on the plane.

$3x - 4y + 2z - 3 = 0$

$3(2+t) - 4(-5+3t) + 2(6+5t) - 3 = 0$

$6 + 3t + 20 - 12t + 12 + 10t - 3 = 0$

$35 + t = 0$

$t = -35$

$\therefore$ a unique t value.

$\Rightarrow$ 1 point of intersection.

$x = -33, y = -110, z = -169$

CASE 3:

$$l: \frac{x+5}{2} = \frac{y}{4}, z=0$$

$$\pi: 2x - y + 4z + 10 = 0$$

Solution:

$$\text{note: } [2, 4, 0] \cdot [2, -1, 4]$$

$$= 0 \Rightarrow \text{parallel}$$

Point $(-5+2t, 4t, 0)$ lies on the plane if

$$2(-5+2t) - 4t + 0 + 10 = 0$$

$$-10 + 4t - 4t + 0 + 10 = 0$$

$$0 = 0$$

$\Rightarrow$ an infinite # of solutions

ie: l lies in the plane.